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Experimental reintroduction of the crayfish species *Orconectes virilis* into formerly acidified Lake 302S (Experimental Lakes Area, Canada)¹

Iain D. Phillips, Rolf D. Vinebrooke, and Michael A. Turner

Abstract: Reintroduction of functionally important species is considered a key strategy for restoring damaged ecosystems. However, the sudden reappearance of an extirpated species may have adverse ecological impacts, degrading ecosystem services. Therefore, we experimentally reintroduced the crayfish *Orconectes virilis* into a biologically recovering, formerly acidified Boreal Shield lake (Lake 302S, Experimental Lakes Area, Canada) to determine its effect on the littoral food web following a 17-year absence. In June 2004, a single-factor experimental design consisting of two treatment levels (crayfish-less control versus 1.8 introduced crayfish·m⁻²) was replicated five times for a total of 10 littoral cages (4 m²). *Orconectes virilis* significantly ($P < 0.05$) suppressed the total biomass of other benthic invertebrates by 70% primarily because of declines in larval damselflies and midges. In addition, crayfish reduced periphytic biomass by 90% ($P < 0.001$). Stable isotopic analyses of the mesocosm food webs further indicated that *O. virilis* likely functioned as an omnivore, exerting direct and possibly indirect effects on other invertebrates and periphyton. Our findings highlight how the reintroduction of *O. virilis* must be balanced with adequate fish predation to prevent this species from becoming an invader and negatively affecting the productive capacity of boreal lakes.

Résumé : La réintroduction d'espèces importantes pour le fonctionnement d'un écosystème est une stratégie essentielle pour la restauration des écosystèmes endommagés. Cependant, le retour subit d'une espèce extirpée peut avoir des impacts écologiques négatifs, ce qui dégrade les services de l'écosystème. C'est pourquoi, nous avons introduit expérimentalement l'écrevisse *Orconectes virilis* dans un lac du Bouclier boréal (lac 302S, Région des lacs expérimentaux, Canada) antérieurement acidifié, mais en phase de récupération biologique, afin d'en déterminer l'effet sur le réseau alimentaire littoral après une absence de 17 ans. En juin 2004, un plan d'expérience à un seul facteur comprenant deux conditions expérimentales (témoin sans écrevisses versus 1,8 écrevisse introduite·m⁻²) a été répété cinq fois dans un ensemble de 10 cages littorales (4 m²). *Orconectes virilis* a réduit significativement ($P < 0,05$) la biomasse totale des autres invertébrés benthiques de 70 %, principalement à cause de la diminution des larves de demoiselles et de moucheron. De plus, les écrevisses ont fait baisser la biomasse du périphyton de 90% ($P < 0,001$). Des analyses des isotopes stables des réseaux alimentaires des mésocosmes ont montré, de plus, qu'*O. virilis* fonctionne comme un omnivore, produisant des effets directs et peut-être aussi indirects sur les autres invertébrés et sur le périphyton. Nos résultats soulignent l'importance de contrebalancer la réintroduction d'*O. virilis* avec une prédation adéquate par les poissons pour empêcher l'espèce de devenir envahissante et d'affecter négativement la capacité de production des lacs boréaux.

[Traduit par la Rédaction]

Introduction

Species interactions are expected to replace abiotic factors as key ecological drivers in ecosystems as they recover from environmental perturbations (Menge and Sutherland 1987; Hogsden and Vinebrooke 2005). In particular, the reestab-

lishment of a major life form is considered a key event in the recovery of perturbed ecosystems, potentially reversing catastrophic state shifts (Scheffer et al. 2001). A common assumption is that functionally important species will recolonize naturally once environmental conditions are amenable to their existence. For example, many benthic arthropods

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are able to rapidly recolonize recovering acidified lakes owing to their aerial dispersal capabilities as adults (Keller et al. 1992; Carbone et al. 1998; Bohonak and Jenkins 2003). However, if dispersal barriers exist, then extirpated species will remain absent despite improving habitat conditions.

Reintroduction of functionally important species is a potential strategy for restoring damaged freshwater ecosystems (International Union for the Conservation of Nature 1998). For example, reintroduction of a common zooplankton species was necessary to restore a historically fishless alpine lake to its natural state following the decline of introduced sportfish (McNaught et al. 1999). However, few other studies have investigated the food web impacts of reintroduced invertebrates into recovering freshwater ecosystems. Nor has much attention been directed to the possibility of reintroduction as a restoration strategy in freshwater ecosystem management outside of proposed crayfish reintroduction strategies in Europe (Taugbøl and Peay 2004).

Crayfish are omnivores that often play key roles in the food webs of lakes and streams (Souty-Grosset 2005), consuming benthic invertebrates, macrophytes, detritus, and periphyton (e.g., Charlebois and Lamberti 1996; Nyström et al. 1996). Unlike obligate top predators that exert strong top-down forces on benthic food chains (Power 1990; Brönmark et al. 1992; Brönmark 1994), omnivores can have more diffuse effects on lower trophic levels, which can confound predictions of trophic cascades (Polis and Holt 1992; Power 1992; Vinebrooke et al. 2001). For example, the food web impacts of several crayfish species, such as *Orconectes virilis* (Dorn and Wojdak 2004), *Orconectes rusticus* (Lodge et al. 1994), and *Pacifastacus leniusculus* (Nyström et al. 1996), have been shown to deviate from the classic trophic cascade model. However, crayfish can also feed selectively depending on prey availability and vulnerability (Nyström et al. 1999). Therefore, reintroduction of *O. virilis* may cause ecological surprises in lakes that are recovering from anthropogenic acidification.

Orconectes virilis is a large omnivorous crayfish sensitive to acidification (Keller et al. 1999). Regional declines of boreal crayfish in northeastern Ontario have been associated with anthropogenic acidification (Errulat et al. 1992; France 1993; Keller et al. 1999). Despite the dramatic reduction in SO₂ emissions since the 1970s, and associated chemical and biological improvements (Gunn and Keller 1990; Keller et al. 1999), *O. virilis* has not recolonized many of these lakes. Further, whole-lake acidification experiments documented the collapse of *O. virilis* as pH declined below 6 (Davies 1989). Lack of reestablishment by many extirpated species, such as *O. virilis*, represents an autecological bottleneck to recovery in acidified boreal lakes that are showing improvements in water chemistry (Keller et al. 1999).

The main goal of our experimental reintroduction of the native crayfish *O. virilis* into enclosures located in a formerly acidified lake was to determine its impact on the littoral food web during chemical recovery. We hypothesized that reintroduction of *O. virilis* at naturally occurring densities would have a restorative effect on the food web of boreal lakes that lack large benthivorous fish, returning them to preacidification conditions (Fig. 1). Specifically, crayfish would selectively suppress large predatory invertebrates, thereby releasing smaller benthic herbivores from

predation, which would then increase grazing pressure and reduce the abundance of periphyton. Both large predatory invertebrates (e.g., larval odonates) and filamentous periphyton proliferate during acidification and persist during chemical recovery until higher trophic levels become reestablished (Schindler et al. 1991; Vinebrooke et al. 2001). The release of grazing pressure on metaphytic green periphyton as *O. virilis* was extirpated may explain, in part, why higher levels of these algae are observed during recovery than during acidification (Turner et al. 2009).

Materials and methods

Study site

The experiment was conducted along the southwest shore of the headwater Lake 302 South (L302S) (49°40'N, 93°45'W) (Fig. 2), which is located in the Experimental Lakes Area (ELA), Ontario, Canada (see Vinebrooke et al. (2001) for a full site description). L302S underwent pH manipulation using H₂SO₄ as part of an acidification experiment (Schindler et al. 1991; Turner et al. 1995a, 1995b) wherein the southern basin was curtained off and its pH reduced from 6.8 to 4.5 during 1982–1991. The native ELA crayfish species *O. virilis* was extirpated from L302S once the pH dropped below 5. The curtains are still in place and together with an outflow impassable to crayfish have prevented the recolonization of L302S by *O. virilis* from downstream Lake 468 (L468). L302S has recovered chemically to a preacidification pH of 6.8, enabling certain periphytic algae, benthic invertebrates, and minnows to become reestablished in the littoral area (Vinebrooke et al. 2001).

Experimental design

A single-factor experimental design consisting of two treatment levels (crayfish-less control versus ~1.8 crayfish-m⁻²) was replicated five times for a total of 10 enclosures (Fig. 2). Enclosures were framed in wood and covered completely with 5 mm mesh sized black aquaculture netting (InterNet Inc., Anoka, Minnesota). The netting would have excluded only fish and late instars of certain large arthropods (e.g., anisopterans). Each cage had a lid and floor or benthic area of 4 m² and was located along the 1 m water depth contour where a predominantly sandy substrate was covered by a thin organic layer and few macrophytes (*Eriocaulon* sp.). The experiment ran for 56 days from 17 June to 12 August 2004.

Cobble baskets ($n = 12$) were introduced randomly into each enclosure to simulate the natural cobble habitats found in L302S and provide discrete sampling units for invertebrates. Each open basket (0.25 m × 0.25 m × 0.15 m) was constructed using 50 mm mesh sized steel hardware cloth. The floor of each basket was reinforced with two pine dowels crossing from corner to corner secured with plastic straps. Eighty-pound test fishing line was used as a handle for the basket strung from the corners and buoyed in the water column with a small styrofoam float. Each basket was filled with cobble collected from L302S (~7–15 cm diameter) and contained four unglazed ceramic tiles (upper surface area = 53.3 cm²) that provided a standard substrate for sampling of periphyton. Cobble baskets and tiles were preconditioned for a 3-week period prior to the experiment

Fig. 1. Hypothesized and experimentally demonstrated impacts of reintroduced *Orconectes virilis* on the food web of a recovering acidified boreal lake (L302S, Experimental Lakes Area, Canada). Arrow thickness intended to convey relative qualitative magnitude of relationships.

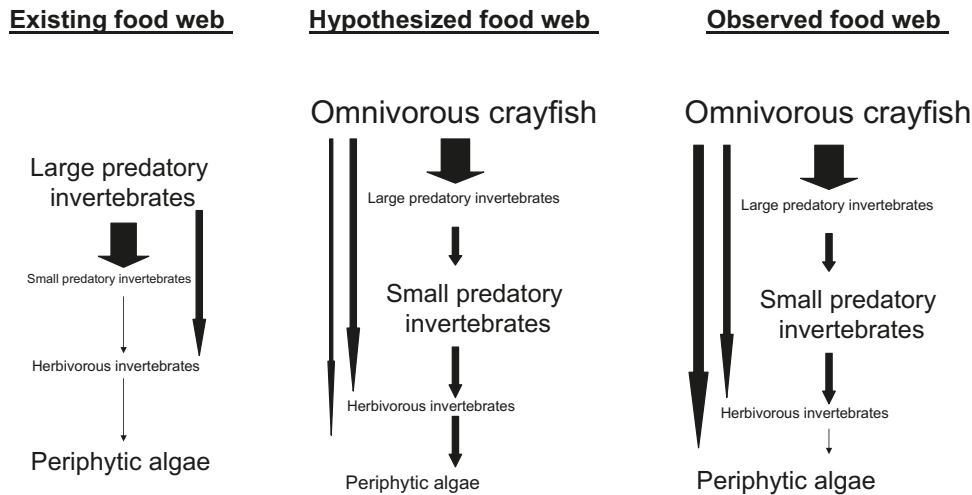


Fig. 2. Study site: experimental *Orconectes virilis* enclosures in L302S, Experimental Lakes Area, Canada.



in the open littoral zone adjacent to the study area to enable colonization by the resident invertebrate and periphyton communities.

Introduction of crayfish

Crayfish were collected from L468 at night by SCUBA divers during 13–16 June 2004. As it is the lake immediately downstream of L302, L468 *O. virilis* populations were likely the most closely related group of crayfish to those in L302 prior to extirpation and the most probable source population for recolonization of L302S in the future. Adult male crayfish (mean carapace length 14.02 ± 1.39 mm (SD) (measured from the anterior apex of the branchiocervical groove to the posterior margin of the carapace)) were intro-

duced into the enclosures on 17 June at a concentration of ~ 1.8 individuals·m⁻² (7 individuals in each enclosure). Only male crayfish were used to prevent an accidental release of a reproductively viable population into the lake. Natural densities of *O. virilis* in other lakes can vary from 1.5–2.6 individuals·m⁻² (Momot and Gowing 1977) to 0.2–0.6 individual·m⁻² (Davies 1989; France and Collins 1993). Further, similar crayfish introduction experiments conducted by Dorn and Mittelbach (2004) and Dorn and Wojdak (2004) employed *O. virilis* concentrations of 1.4 and 1.5 individuals·m⁻².

Sampling of zoobenthos

At 2-week intervals, three cobble baskets were randomly

selected from each enclosure and sampled for periphyton and zoobenthos. Approximately 50% of the shallow littoral habitat in L302S is cobble-based, while the rest comprises mainly granite outcroppings and natural sandy beach; therefore, cobble baskets represented well the most common, as well as preferred, habitat type for this crayfish species. Tiles were placed carefully into a sealed container underwater to prevent physical disturbance prior to removal of the basket. Cobble baskets were slowly relocated into a plastic container and then lifted to the surface. As the basket was in an enclosed container when removed, we are confident that invertebrate losses due to basket removal were minimal or at least consistent across both treatments.

Cobble was scrubbed by hand to dislodge invertebrates, which were then concentrated onto a 90 μm sieve. Large predatory invertebrates (e.g., Odonata) were then removed from the sieve and samples placed on ice to be sorted in the laboratory within 5 days. Invertebrates were identified to the lowest possible taxon under a dissecting microscope using taxonomic keys (Davies 1971; Clifford 1991; Merritt and Cummins 1996; Wiggins 1996); their body length was measured to the nearest 1 mm. All zoobenthos samples were frozen immediately after sorting. Intact invertebrates were freeze-dried and weighed to determine dry weights. Weights of partially damaged individuals were calculated from the body length – dry weight conversion equations of Benke et al. (1999). Most invertebrates were in adequate condition for individual weighing, and equations were used rarely. Data were expressed as the sum of the three cobble baskets for each enclosure.

Sampling of periphyton and leaf detritus

On each sampling date, all four tiles from each sampled cobble basket were pooled and periphyton was harvested using a toothbrush and then filtered through a Whatman GF/C filter (pore size = 1.2 μm). Filters were frozen immediately. Biomass was quantified using ash-free dry weight as a proxy. The filters were freeze-dried, periphyton scraped off and homogenized with a mortar and pestle. Ash-free dry weight was then determined for each sampling day by combusting a subsample and calculating differences between dry and ash (at 500 °C for 24 h) weights. Periphyton pigment concentrations were quantified using a reversed-phase, high-pressure liquid chromatography procedure (Vinebrooke and Leavitt 1999). Pigments were extracted by soaking freeze-dried periphyton subsamples in acetone, methanol, and water (80:15:5 v/v) for 24 h in darkness at –10 °C. Extracts were filtered through 0.2 μm pore nylon filters, dried, and stored under nitrogen gas in the dark at –10 °C. Dried extracts were reconstituted using an injection solution (70% acetone, 25% ion-pairing reagent, and 5% methanol). Pigments were separated on an Agilent (Hewlett-Packard Canada Ltd., Mississauga, Ontario, Canada) model 1100 system with a Varian Microsorb model 100 column (10 cm length, 3 μm particle size). Pigments were detected with an in-line HP Series 1100 diode array detector (435 nm detection wavelength) and a fluorescence detector (435 nm excitation wavelength, 667 nm detection wavelength).

Fallen leaves were collected from the riparian zone around L302S prior to the launch of the experiment. Paper birch (*Betula papyrifera*) and alder (*Alnus* sp.) leaves were

separated from the leaf litter and air-dried to a constant weight. Leaves were enclosed in loose mesh bags (25 mm mesh size) and weighed to provide leaf packs of 23 ± 4 g available to crayfish. Five leaf packs were incubated in each enclosure and attached to cobble baskets with plastic cable ties. All leaf packs were sampled at day 56, at the conclusion of the experiment. They were removed and oven-dried at 50 °C to a constant weight and then reweighed.

Crayfish recovery and gut analysis

At the termination of the experiment, remaining crayfish were collected by active search and removed from the enclosures. Only a few crayfish (two to four) in each enclosure were not retrieved at the end of the experiment. Trap sampling for crayfish in L302S during the summer of 2004 (720 trap-days, June–August) is consistent with the contention that there are still no “wild” *O. virilis* present (i.e., “losses” during the experiment likely represented mortality).

All crayfish recovered from the enclosures were dissected and their foreguts removed. Only contents of the foregut were examined, as nearly all material in the mid- and hindguts was digested beyond recognition. The foregut of each crayfish was excised and its contents emptied into a water-filled 1 mm grid lined Petri dish, stirred to distribute evenly, and viewed with a dissecting microscope; however, for the most part, contents were not identifiable. What gut contents were found, however, were classified as detritus, algae, or invertebrate. Percent total area of the dish covered by each classification was estimated using a 1 mm² grid. Subsamples were taken and examined with a compound microscope to confirm the identity of items. All invertebrate individuals were identified to the most specific level possible. Immediately after dissection, the remaining crayfish material was refrozen for stable isotope analysis. At the end of the experiment, 23 adult male crayfish of comparable size to the experimental crayfish were collected from the source lake (L468) and analyzed for gut contents and stable isotopes to identify how closely the diets of experimental crayfish were to local/source populations.

Stable isotope analyses

The combination of stable isotopes and gut content analysis can provide reliable quantitative estimates of the contributions of energy sources to production of crayfish and other aquatic organisms. Integrating C and N stable isotope elements allows identification of the original C source for a community (Hecky and Hesslein 1995) and evaluation of the trophic relationship among organisms using N (Hobson and Welch 1992). Whitledge and Rabeni (1997) found that crayfish dietary trophic position calculated from gut content analysis correlates well with their trophic position determined from tissue $\delta^{15}\text{N}$. However, whereas gut analysis only provides a snapshot of what an organism consumed immediately prior to capture, stable isotope analyses integrate a longer timeframe of dietary consumption.

To obtain the biomass required for the analyses, taxa were pooled into their respective families and across the 10 replicates for each sampling date. Chironomids were separated into tanypodine and nontanypodine chironomids, as Tanypodinae is the only taxon in Chironomidae that is predomi-

nantly carnivorous (Merritt and Cummins 1996) and their trophic function may have differed from that of other chironomids in the enclosures. Although the variance of isotopic composition across replicates can no longer be determined when one pools the organisms, mean values can still be recorded (Rosenfeld and Roff 1992; Nyström et al. 1999). Stable isotope analysis was also conducted on abdominal muscle tissue (France 1996) of all crayfish recovered from the enclosures and crayfish from the source population. All samples were homogenized by grinding into a fine powder with a mortar and pestle. Isotopic analysis was performed at the Biogeochemical Analysis Laboratory, University of Alberta (Edmonton, Alberta).

Statistical analyses

Repeated-measures analyses of variance (RM-ANOVAs) and RM multivariate analyses of variance (RM-MANOVAs) were conducted to test for the time-dependent effects of crayfish reintroduction on total community biomass and taxonomic composition, respectively. Significant RM-MANOVA results were further examined using individual RM-ANOVAs to identify the effects of crayfish on individual zoobenthos taxa and taxonomically diagnostic algal pigments. Analyses were performed using SPSS software version 11.0.3 (SPSS Inc., Chicago, Illinois).

Results

Crayfish impacts on consumers

Crayfish consistently suppressed total invertebrate biomass (RM-ANOVA, treatment effect, $P < 0.05$) (Table 1; Fig. 3) and altered taxonomic composition (RM-MANOVA, treatment effect, $P = 0.003$) during the experiment. RM-ANOVAs revealed that the negative impact of crayfish on zoobenthos biomass was attributable to effects on odonates, leeches, and chironomids (Table 1; Fig. 4). In contrast, crayfish had no significant effect on the biomass of ephemeropterans or trichopterans (Fig. 4).

Crayfish treatments had contrasting species-specific effects on odonates (Fig. 4; Table 1). *Orconectes virilis* enclosures contained significantly less biomass of the most abundant odonate, *Aeshna* sp. However, crayfish significantly increased the population density of this anisopteran (RM-ANOVA, $F = 4.37$, $P = 0.04$) (Fig. 5). In contrast, crayfish significantly increased total zygopteran biomass, consisting primarily of *Enallagma* sp. (Table 1; Fig. 4).

Crayfish impacts on periphyton and leaf litter

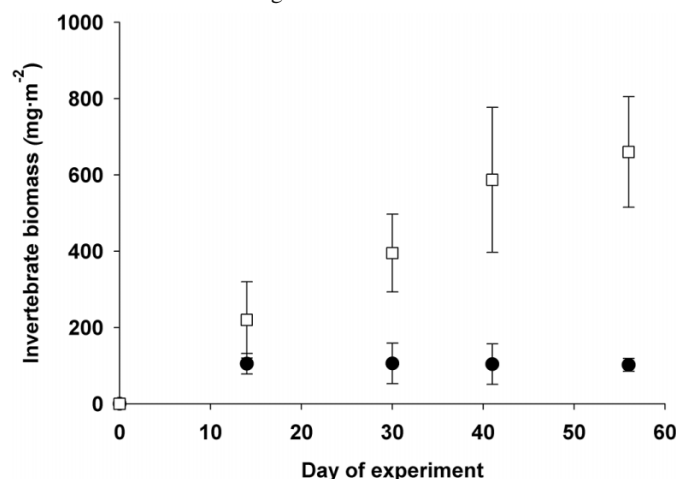
The presence of crayfish significantly decreased ash-free dry weight inferred total periphyton biomass (Table 1; Fig. 6) and altered the taxonomic composition of the algal community (RM-MANOVA, treatment $\times$ time interaction, $P = 0.03$). Specifically, RM-ANOVAs revealed that *O. virilis* reduced chlorophyll concentrations and altered taxonomic composition through disproportionately suppressing filamentous green algae containing the algal pigments chlorophyll *b* and lutein (Table 2; Fig. 7). Similarly, crayfish significantly decreased benthic diatoms containing fucoxanthin and diatoxanthin. In contrast, crayfish did not significantly (Student *t* test, $P > 0.05$) affect the final total biomass of litter packs.

Table 1. Repeated-measures ANOVA results for the effects of *Orconectes virilis* introductions on periphyton and benthic invertebrate biomasses.

Source	Treatment	Time	Treatment $\times$ time
Periphyton	117.72**	903.2**	13.918**
Total invertebrates	5.66*	1.58	0.81
Chironomidae (Diptera)	14.90**	14.97**	2.27
<i>Caenis</i> (Ephemeroptera)	0.489	26.64**	0.96
<i>Oecetis</i> (Trichoptera)	0.269	3.83*	0.548
<i>Aeshna</i> (Anisoptera)	11.49**	5.6**	2.6
<i>Enallagma</i> (Zygoptera)	6.35*	2.99	0.63
Hirudinea	11.79**	0.94	0.94

Note: Values are *F* statistics for $n = 5$ replicates: * $P < 0.05$ and ** $P < 0.001$.

Fig. 3. Mean (± 1 SE) biomass of total benthic invertebrates in *Orconectes virilis* enclosures (circles) and control enclosures (squares) in L302S from June to August 2004.



Stable isotopic and gut analyses

Introduced crayfish in L302S and the source population from L468 occupied top trophic positions close to the secondary consumers based on their $\delta^{15}\text{N}$ signatures and a standard enrichment of 3.4‰ per trophic level (Fig. 8). In comparison, predatory leeches (*Nepheleopsis obscura*) had slightly enriched $\delta^{15}\text{N}$ values, while values were similar among the odonates *Aeshna* sp. and *Enallagma* sp. and the crayfish. Other invertebrates that also occupied the same intermediate trophic position were the predatory tanypodine chironomids and trichopterans. Primary consumers included herbivorous ephemeropterans and nontanypodine chironomids.

Periphyton was the primary C source to most benthic invertebrates based on their well-aligned $\delta^{13}\text{C}$ values (Fig. 8). In contrast, leaf detritus $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ was too far removed from the rest of the food web likely be of any significant food source for *O. virilis*. Crayfish from the enclosures and L468 were enriched in $\delta^{13}\text{C}$ by ~3‰ relative to the rest of the food web. Macrophytes would have increased the potential array of prey items for crayfish (e.g., plant matter, epiphytic invertebrates) but were not observed to occur in the littoral zone of L302S. Also, assimilation of macrophyte ma-

Fig. 4. Mean (± 1 SE) biomass of (a) *Aeshna*, (b) Ephemeroptera, (c) *Nephelopsis obscura*, (d) *Enallagma*, (e) Trichoptera, and (f) Chironomidae benthic macroinvertebrates in *Orconectes virilis* enclosures (circles) and control enclosures (squares) in L302S from June to August 2004.

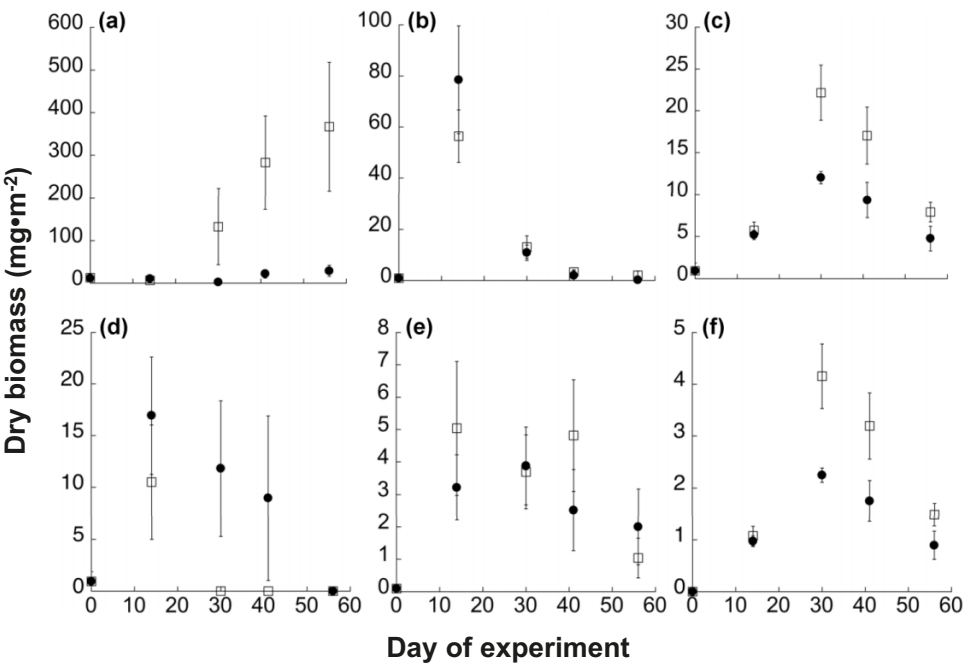


Fig. 5. Mean (± 1 SE) abundance of the dominant benthic predator *Aeshna* in *Orconectes virilis* enclosures (circles) and control enclosures (squares) in L302S from June to August 2004.

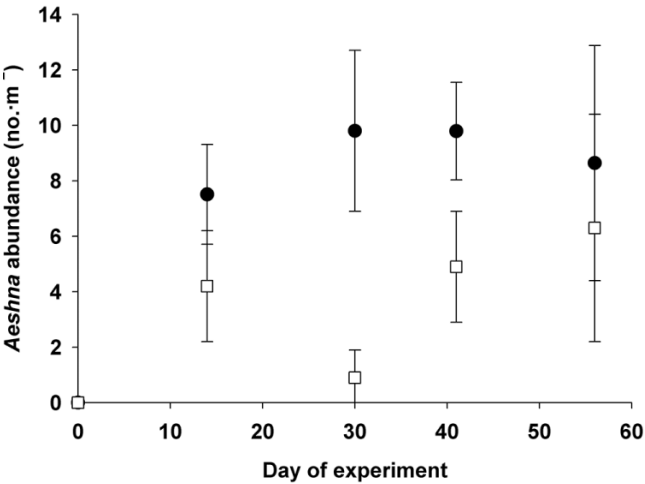


Fig. 6. Mean (± 1 SE) biomass of periphyton in *Orconectes virilis* enclosures (circles) and control enclosures (squares) in L302S from June to August 2004.

terial may have reduced the isotopically inferred trophic position of the crayfish, making it appear more herbivorous.

As measurements were made at the termination of the experiment, too much time lapsed and what material was present in crayfish guts was highly digested and too difficult to identify accurately. However, some detrital material, sand, and wood fragments were observed in the samples. Further, two crayfish from one enclosure had large amounts of crayfish carapace in their fore gut that likely originated from a freshly molted individual in this group but could also have resulted from cannibalism on the missing crayfish aforementioned.

Table 2. Repeated-measures ANOVA results showing the effects of *Orconectes virilis* introductions on periphyton pigment concentrations.

Source	Treatment	Time	Treatment $\times$ time
Fucoxanthin	69.43**	2.38	2.013
Diatoxanthin	49.00**	13.54**	1.36
Lutein/zeaxanthin	31.63**	4.84**	2.36
Chlorophyll <i>a</i>	59.60**	3.88*	3.65*
Chlorophyll <i>b</i>	44.46**	9.74**	3.709*

Note: Values are *F* statistics for *n* = 5 replicates: **P* < 0.05 and ***P* < 0.001.

Fig. 7. Mean (± 1 SE) concentration of the algal pigments (a) chlorophyll *a*, (b) chlorophyll *b*, (c) fucoxanthin, (d) lutein/zeaxanthin, and (e) diatoxanthin in *Orconectes virilis* enclosures (circles) and control enclosures (squares) in L302S from June to August 2004.

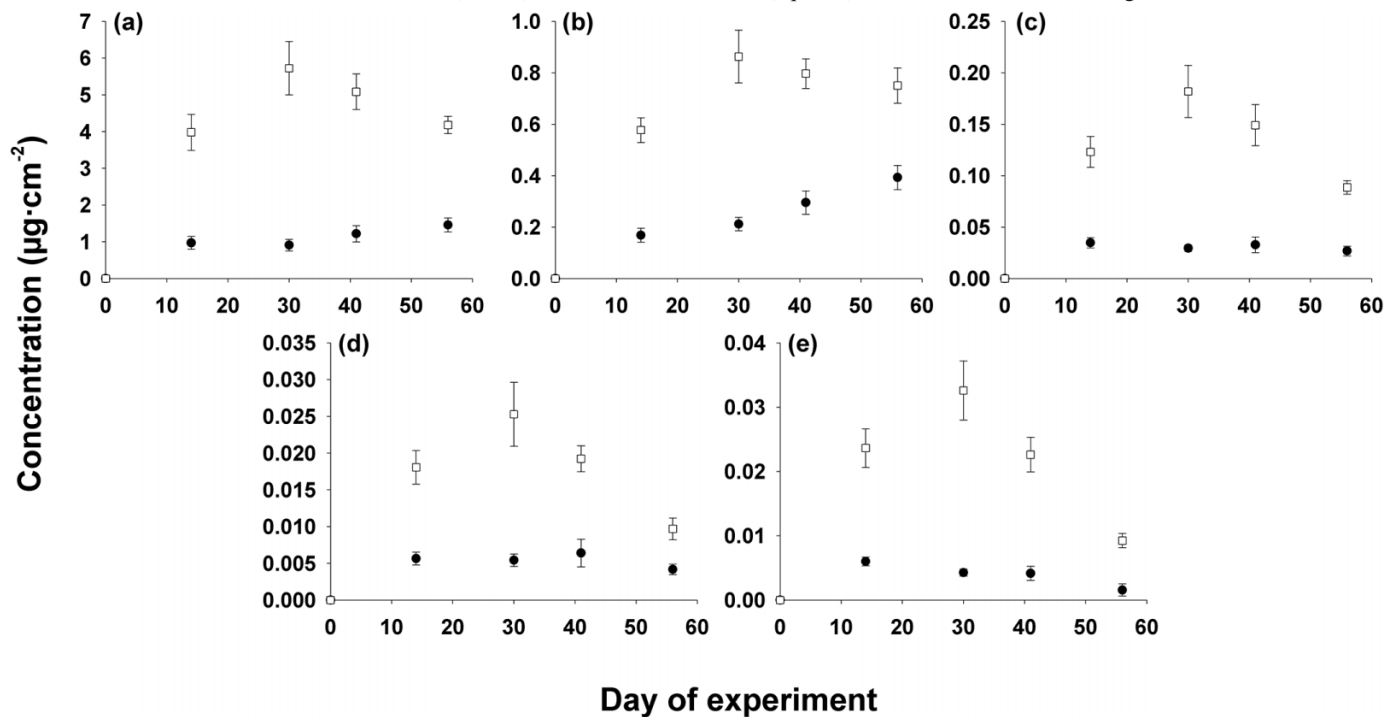
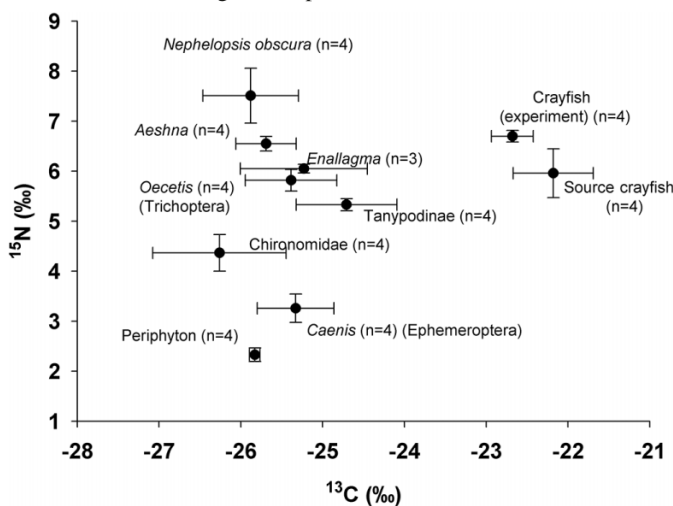


Fig. 8. Results of stable isotope analyses showing mean (± 1 SE) $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ of benthic fauna from summer 2004 in L302S. Number of replicate samples used to determine the isotopic signature of each taxon is given in parentheses.



Discussion

Our findings provided partial support for the hypothesized food web impacts of reintroduced crayfish in lakes recovering from acidification. Crayfish significantly reduced total zoobenthos biomass by selectively suppressing large predatory odonates. However, smaller predatory invertebrates benefited from the presence of crayfish, while the biomass of herbivorous invertebrates was unaffected. Further, the presence of crayfish unexpectedly suppressed, rather than indirectly stimulated, periphytic biomass. Therefore, we must refine our hypothesized food web effects to account for the

more complex omnivorous impact of *O. virilis*. Below, we consider both possible ecological mechanisms underlying observed food web impacts of crayfish and their potential ramifications for restoration of damaged lake ecosystems.

Crayfish impacts on consumers

Our findings support other reports of crayfish functioning as strong benthivores in freshwater lakes (Hanson et al. 1990; Lodge et al. 1994; McCarthy et al. 2006) and streams (Helms and Creed 2005). In particular, *O. virilis* strongly suppressed the total abundance of the anisopteran *Aeshna* sp. by negatively affecting larger individuals belonging to this genus. However, densities of *Aeshna* sp. were significantly elevated in the enclosures relative to the crayfish-less controls because of the increase in small individuals. Similarly, the smaller zygopteran genus *Enallagma* sp. was also more abundant in the presence of crayfish. Therefore, we suggest that size-selective predation by crayfish on larger odonates, or possibly their active avoidance of this large predator, released smaller odonates from competition and cannibalism (Stoks and McPeck 2003a, 2003b).

Stenroth et al. (2006) have suggested that tissue turnover rates in crayfish may be slow enough to cover differences in diet between months. Hence, it is possible that stable isotope evidence of direct crayfish predation may have been masked by the previous dietary habits of the crayfish in their source lake. Despite this potential limitation, we can reconstruct the rest of the food web present in the enclosures with stable isotopes. When used in combination with the demonstrated effects of crayfish on prey abundances, we can develop a reliable picture of crayfish reintroduction effects on benthic communities. Evidence from stable isotope analyses suggests that active avoidance rather than predation may better

explain why crayfish suppressed large odonates in the enclosures. Crayfish did not have a highly elevated $\delta^{15}\text{N}$ signal relative to odonates, suggesting that direct consumption may not have been prominent. However, we did not process odonates from the source lake for stable isotopic analyses to help support this assertion or collect gut contents from the earlier stages of the experiment. As mentioned above, it is possible that the crayfish did not have adequate time in the experiment to express their enclosure diet in their tissue, although the crayfish $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signals moved slightly toward that of the new food web.

Nevertheless, few physical refuges for large odonates from predation would be available given a whole-lake reintroduction of crayfish. Further, it is likely that our results reflect a lack of predation on younger *Aeshna* sp. instars because of their ability to access refuges too small for the adult crayfish used here. If we had used a full size range of crayfish in our experiment, then we may have found crayfish predation on all *Aeshna* sp. life stages; thus, whole-lake reintroduction may have dramatic consequences on the dragonfly population.

Johnson (1991) described the necessity of refugia for odonates in their efforts to find a balance between active foraging and hiding from predators. An alternative possibility is that crayfish excluded large anisopterans from decent refugia, thereby increasing their susceptibility to predation by resident minnows (Vinebrooke et al. 2001). Further, Kirk and Smock (2000) observed that competition for physical refugia was intense among crayfish and large predacious invertebrates, while crayfish often shared these spaces with smaller individuals of the same invertebrate species. Therefore, crayfish likely reduced large odonates from the enclosures owing to a combination of predation and competition. Similarly, competitive interference could also account for why crayfish significantly suppressed leeches, which occupied a similar trophic position as *O. virilis* based on stable isotopic evidence.

A potential explanation for the observed concomitant decreases in large instars of the odonate *Aeshna* sp. and increases in younger instars of *Aeshna* sp. and smaller *Enallagma* sp. is release from predation and competition owing to size-selective feeding by crayfish. Large odonate larvae have the potential to outcompete small larvae for prey, or quite simply prey on smaller larvae (Benke et al. 1982; Morin 1984). Importantly, the closely related invasive rusty crayfish *O. rusticus* exerted the opposite negative effect on *Enallagma* (McCarthy et al. 2006) by eliminating its preferred macrophytic habitat (Dionne and Folt 1991; Hilsenhoff 1995). These species-specific effects of *Orconectes* highlight how native and exotic congeners can exert contrasting impacts in freshwater ecosystems.

Our hypothesized direct link between *O. virilis* and the herbivores was not realized. Although *O. virilis* significantly suppressed chironomids, they did not exert impacts on other herbivores, such as the Ephemeroptera. It is possible that release from predation by large anisopterans offset the negative effect of crayfish predation on the ephemeropterans. However, we cannot exclude that the possibility that predation pressure by odonates remained constant owing to high densities of smaller species compensating for reduced abundance of larger species in the presence of crayfish.

Crayfish impacts on producers

Orconectes virilis were observed nonselectively consuming periphyton during the experiment (I.D. Phillips, personal observation), and sources of macrophytes or detritus outside the leaf packs were sparse. Macrophytes in particular were not observed in the enclosures, nor were they abundant in the surrounding littoral zone of the lake.

The direct negative impact of crayfish on periphytic biomass likely involved the combined effects of grazing and physical disturbance while this omnivore foraged for invertebrate prey. Gut analysis revealed that crayfish consumed detritus containing bacteria and several algal groups. Therefore, crayfish suppressed periphytic algae partly by direct consumption of loosely attached species. These findings agree well with the findings of Keller et al. (1999) who found significantly higher periphytic biomass in crayfishless lakes recovering from anthropogenic acidification in northeastern Ontario. However, firmly attached filamentous green algae (chlorophyll *b*) increased steadily in the presence of crayfish, indicating the relative inefficiency of crayfish as grazers when compared with other benthic invertebrates, such as ephemeropterans and trichopterans (Steinman 1996).

Unlike the herbivorous zoobenthos, omnivorous crayfish exhibited a more enriched $\delta^{13}\text{C}$ signal relative to periphyton. The $\delta^{13}\text{C}$ enrichment may have been a result of preferential assimilation of certain components of organic matter that do not reflect its bulk $\delta^{13}\text{C}$ signal (McCutchan et al. 2003). However, it is more likely that the length of this experiment was insufficient to allow for tissue turnover and detection of *O. virilis* diet in the enclosures (Stenroth et al. 2006). Thus, the difference between crayfish and periphytic $\delta^{13}\text{C}$ signatures may reflect consumption of unmeasured alternate sources of organic C in the source lake (L468), such as $\delta^{13}\text{C}$ -enriched epilithic algae (France and Peters 1995). Further, Stenroth et al. (2006) also suggested that omnivorous crayfish may decouple their sources of N and C, consuming invertebrates to meet their N demand and plant material for their C budget. Although the benthic communities were likely similar among the enclosures in L302S and the nearby downstream L468, crayfish diet may have varied because of differences in certain habitat conditions, such as fish predation affecting feeding behavior in L468.

Management implications

Reintroduction has been specified by the International Union for the Conservation of Nature (1998) Reintroduction Specialist Group as a key strategy for reestablishing extirpated species to their native home range. Species reintroductions are often performed in terrestrial ecosystems (see Clark et al. 2002), and the results have been mixed. Such reintroductions have been costly, time consuming, and only ~11% successful (Earnhardt 1999); however, the recommendations arising from these experiments appear compatible with the nature of crayfish populations. For example, reintroduction success is enhanced with large founding populations, low environmental variation, high rate of population increase, and low intraspecific competition (Griffith et al. 1989). In addition, lake ecosystems lack the dramatic environmental variation experienced by terrestrial ecosystems that may otherwise compromise fragile founding populations. Fortu-

nately, crayfish reintroductions in boreal ecosystems would benefit from large source populations of *O. virilis* being readily available in surrounding areas. Further, with pH returned to acceptable levels for *O. virilis* inhabitation and reproduction (>5.9; France 1993), female *O. virilis* can theoretically produce >250 offspring each year (Corey 1987) and when entering a lake without conspecifics or resident crayfish *O. virilis* will likely proliferate quickly.

O. rconectes virilis is not a species that is in danger of extinction, as its range and population are much larger than those of European crayfish. Further, reintroductions in Europe have different biodiversity and socioeconomic rationales than here in North America. Crayfish resources are important in European diets (Swahn 2004), and these resources were seriously compromised owing to the crayfish plague (*Aphanomyces astaci*) and the invasion of plague-carrying crayfish species (Taugbøl and Skrudal 1999; Vogt 1999).

Organizations such as the European network CRAYNET are discussing reintroduction of indigenous crayfish and habitat restoration as a method to restore populations recently lost to their historic range (Souty-Grosset 2005). Managers in Norway have already successfully reintroduced noble crayfish (*Astacus astacus*) to the Glomma and Halden watercourses, and guidelines for future reintroductions are being developed (Taugbøl and Peay 2004).

In North America, crayfish reintroduction to previously acidified waterbodies may be an essential step towards ecosystem restoration. As long-range acid deposition has affected large-scale regions, it is likely that significant physical barriers exist that will restrict native populations of *O. virilis* from returning to their native range. Therefore, if aquatic resource managers are interested in returning aquatic ecosystems to their preacidification conditions, then it is necessary to reintroduce a keystone species such as *O. virilis*.

However, restocking of previously acidified lakes with native crayfish in the absence of large fish may block energy transfer to higher trophic levels, possibly decoupling benthic from pelagic habitats. Moreover, reintroduction of *O. virilis* may express ecologically detrimental invader-like traits in the absence of suitable predators. Mack et al. (2000) asserted that a variety of traits define invaders and the impact they cause. If the mere presence of a new species alters fundamental ecological properties such as the dominant species in a community and the abiotic features of an ecosystem, plant productivity, and nutrient cycling, then an invasive species will be detrimental.

We have shown that *O. virilis* has a dramatic effect on the benthic environment by replacing previously top benthic predators (leeches and odonates), reducing periphytic total and algal biomass. Consequently, reintroduced *O. virilis* may affect a trophic deadend in lakes lacking predators. Alternately, crayfish foraging among periphytic detritus may facilitate nutrient recycling, promoting littoral productivity. If the goal of postacidification lake restoration is to return to preacidification conditions, then reintroduction is a necessary step in that direction. However, *O. virilis* reintroductions may degrade ecosystem qualities we hold as virtues by redirecting energy flow in a recovering lake. Therefore, reintroduction of *O. virilis* will need to be considered in the

context of the fish present to avoid potential competitive interference with forage fish (e.g., cyprinid minnows; Vinebrooke et al. 2001) and major harvestable fish species that are incapable of predatory suppression of crayfish.

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